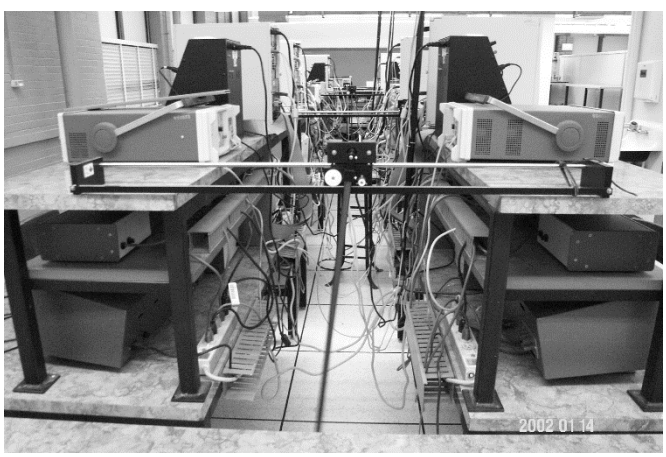
# HOMEWORK SET 2

**Modeling, Identification and Control of Mechatronic Systems** **Fall 2022**
OUT: 26/09/22 DUE: 09/10/22

Use separate sheets to answer each problem with clear problem number, figures, equations and illustrations. **Please submit the answer sheets together with Matlab codes and Simulink model (Matlab2015a Version or below)** used in your problem analysis when you hand in your homework.

**PROBLEM #1**(100%)

The gantry-crane system is shown in Fig. 1.1 and Fig.1.2, in which $x_c$ represents the position of the cart, $F_m$ represents the control force that will be applied to the cart by the motor (attached on the cart), $\theta$ represents the rotational angle of the rod with respect to vertical axis, and $x_p$ is the tip position of the rod. The mechanical system in Fig.1.1 consists of a cart moving on a track with a rod attached to the cart body through a rotary joint with little rotary damping. Let $m_c$ be the mass of the cart, $b_c$ the coefficient of the viscous damping friction between the cart and the track, $F_c$ the Coulomb friction between the cart and the track, $m_r$ the mass of rod, $L$ the length of the rod, $I$ the moment of inertia of the rod with respect to its center of gravity, and $b_r$ the damping coefficient of friction torque at the rotary joint between the rod and the cart. For the actual system set-up, the control force $F_m$ in Problem 1 is the transmitted force to the cart through a gear set powered by a DC motor. The parameters of the system are given in Table 1.



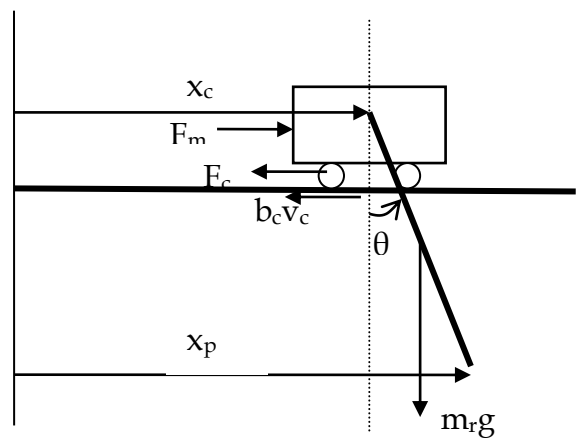
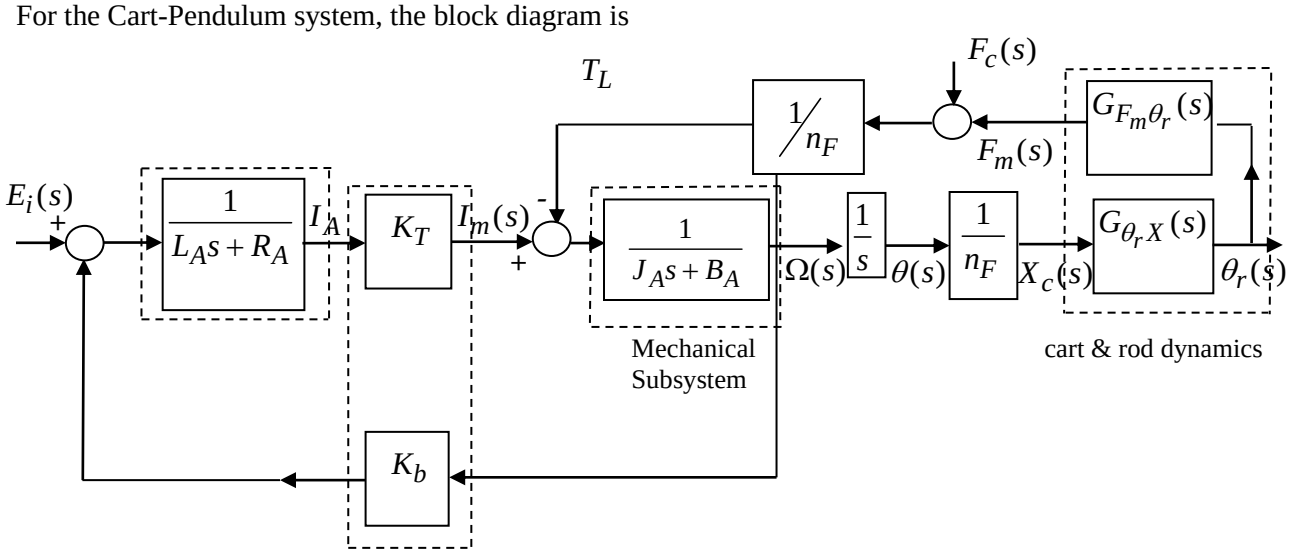




Fig.1.1 Gantry-Crane System Figure 1.2: Sketch of Gantry Crane Systems

For the Cart-Pendulum system, the block diagram is



The transfer function from $X_c$ to $\theta_r$ is

$$G_{\theta_r X}(s) = \frac{\theta_r(s)}{X_c(s)} = \frac{-\frac{1}{2}m_r L s^2}{(I + \frac{1}{4}m_r L^2)s^2 + b_r s + \frac{1}{2}m_r g L} \tag{1}$$

The physical parameters (i.e., $m_r$, $I$, $L$, $b_r$ , $J_A$, $B_A$ etc.) of Eq.(1) can be written in form of lumped parameters (i.e, $c_0$,$c_1$,$d_0$):

$$G_{\theta_r X}(s) = \frac{d_0 s^2}{s^2 + c_1 s + c_0} \tag{2}$$

Where $c_0 = \dfrac{\frac{1}{2}m_r g L}{I + \frac{1}{4}m_r L^2}$, $c_1 = \dfrac{b_r}{I + \frac{1}{4}m_r L^2}$, $d_0 = -\dfrac{\frac{1}{2}m_r L}{I + \frac{1}{4}m_r L^2}$

On the other hand, the transfer function from $\theta_r$ to $F_m$ is

$$G_{F_m \theta_r}(s) = \left[\frac{(m_r + m_c)s^2 + b_c s}{G_{\theta_r X}(s)} + \frac{1}{2}m_r L s^2\right] \tag{3}$$

Thus, the transfer function from control voltage $E$ to $\theta_r$ to is

$$G_{\theta_r E}(s) = \frac{-\frac{1}{2}n_F K_T m_r L\, s}{a_4 s^4 + a_3 s^3 + a_2 s^2 + a_1 s + a_0} \tag{4}$$

And the transfer function from friction force $F_c$ to $\theta_r$ is

$$G_{\theta_r F_c}(s) = \frac{\frac{1}{2}m_r L\left(L_A s + R_A\right)s}{a_4 s^4 + a_3 s^3 + a_2 s^2 + a_1 s + a_0} \tag{5}$$

where the coefficients of the denominator are given by

$$a_4 = n_F{}^2 J_A L_A\left(I + \frac{1}{4}m_r L^2\right) + L_A\left[(m_r + m_c)(I + \frac{1}{4}m_r L^2) - \frac{1}{4}m_r{}^2 L^2\right],$$

$$a_3 = n_F{}^2\left[J_A L_A b_r + (b_0 L_A + R_A J_A)\left(I + \frac{1}{4}m_r L^2\right)\right] + L_A\left[(m_r + m_c)b_r + b_c(I + \frac{1}{4}m_r L^2)\right] + R_A\left[(m_r + m_c)(I + \frac{1}{4}m_r L^2) - \frac{1}{4}m_r{}^2 L^2\right],$$

$$a_2 = n_F{}^2\left[\frac{1}{2}m_r g L J_A L_A + (b_0 L_A + R_A J_A)b_r + (R_A b_0 + K_T K_b)\left(I + \frac{1}{4}m_r L^2\right)\right] + L_A\left[\frac{1}{2}m_r g L(m_r + m_c) + b_c b_r\right] + R_A\left[(m_r + m_c)b_r + b_c(I + \frac{1}{4}m_r L^2)\right],$$

$$a_1 = n_F{}^2\left[\frac{1}{2}m_r g L(b_0 L_A + R_A J_A) + b_r(R_A b_0 + K_T K_b)\right] + \frac{1}{2}m_r g L b_c L_A + R_A\left[\frac{1}{2}m_r g L(m_r + m_c) + b_r b_c\right]$$

$$a_0 = \frac{1}{2}m_r g L n_F{}^2(R_A b_0 + K_T K_b) + \frac{1}{2}m_r g L b_c R_A$$

The measurement signals are the cart position $X_c$ and the rod angle $\theta_r$. The control input $E$ is the applied voltage of DC motor. We will try system identification to obtain several unknown lumped parameters of the system. To do this, the following procedures are considered.

(1) Please conduct system identification **in time domain** or **frequency domain** to obtain the unknown lumped parameters. You can use system identification toolbox in Matlab (command "ident") to help you accomplish the work. Please describe the relevant techniques with necessary figures and illustrations, such as **generating regressor models in time domain** or **process models in frequency domain** from dynamics, **generating input signals**, **selecting useful data for system identification**, **doing parameter estimation, verifying results of system identification** etc. *Note that a stable filter function with relative order more than* 1 *could be applied into both sides of dynamic equation in order to achieve effective signals for system identification. Also note that the structure of the process model can be estimated through linearization of the dynamics nearby initial status.*

(2) Suppose the friction force is zero, please check whether the estimated parameters will approach to their true values. If the friction force is not zero, the estimated parameters will also approach to their true values? Please try the system identification under different friction force disturbances and comment the results you observed.

**Table** 1 Physical Parameters of the Gantry-Crane System

| System | | Variable | Physical Meaning | Value | Unit |
|---|---|---|---|---|---|
| **CART** | | $m_c$ | Mass of Cart + Additional Mass | 0.911 | kg |
| | | $b_c$ | Damping Coefficient of Cart due to the viscous friction between the cart and the track | 5.2 | kg/sec |
| | | $F_c$ | Nonlinear Coulomb friction between the cart and the track | | |
| | | $F_m$ | Applied control force on the cart from the motor | | |
| **ROD** | | $m_r$ | Mass of Rod | 0.231 | kg |
| | | $L$ | Length of rod | 0.64 | m |
| | | $g$ | Gravitational acceleration | 9.81 | $ms^{-2}$ |
| | | $I$ | Moment of Inertia of the rod with respect to the center of its gravity | $I$=1/12 $m_rL^2$ | |
| | | $b_r$ | Rotary damping coefficient at the rotary joint between the rod and the cart | 0.0015 | kg/sec |
| **DC motor** | | $K_T$ | Motor torque constant | 0.00767 | Nm/amp |
| | | $K_b$ | Back EMF constant | 0.00767 | V/(rad/sec) |
| | | $R_a$ | Armature resistance | 2.6 | Ω |
| | | $L_a$ | Armature inductance | 0.18 | mH |
| | | $n$ | Internal gear ratio | 3.7:1 | |
| | | $r$ | Radius of the motor pinion that meshes with the track | 0.00635 | m |
| | | $J_0$ | Armature inertia | 3.87e-7 | $Kg.m^2$ |
| | | $b_0$ | Motor and gear bearing viscous friction coefficient | 0 | Kg/sec |